

Controlled Class-Imbalance Construction for TCGA-UT

Yannik Queisler

June 12, 2026

Abstract

Class imbalance in computational pathology is not only a question of how many examples a class contributes. A class may perform poorly because it is rare, because its frozen foundation-model features overlap with other classes, or because both effects coincide. The companion TCGA-UT imbalance benchmark established the dataset, the mitigation taxonomy, and patch-level and WSI-bag evaluations under the native TCGA-UT long tail. This report defines a controlled counterpart: artificial full-scale TCGA-UT training distributions that keep all 32 cancer types, preserve fixed validation and test splits, and vary only the training support assigned to each class. Power-law training distributions are generated under native-prevalence, easy-to-difficult, and difficult-to-easy class orders, using frozen Virchow2 patch features and WSI-bag features with the same 30-instance evidence budget as the companion benchmark. The design separates imbalance severity from the identity of the classes assigned to the head and tail of the training distribution. Discrimination and calibration are evaluated for patch-feature and WSI-bag mitigation methods in their native regimes, including post-hoc temperature scaling fitted on validation predictions.

Contents

1	Introduction	2
2	Dataset Construction	2
2.1	Full-Scale Training Pool	2
2.2	Power-Law Training Support	3
2.3	Class Order	3
3	Methods	3
4	Evaluation Protocol	4
5	Results	4
6	Discussion	6
7	Conclusion	6

1 Introduction

The companion TCGA-UT imbalance benchmark compared class-imbalance mitigation methods in their native input regimes: labeled patch-feature classification and weakly supervised WSI-bag classification. It also showed that class frequency alone is an incomplete explanation for poor classwise performance. Some low-support cancer types remain separable under a frozen pathology foundation model, whereas other classes are difficult even when their support is not the smallest. This report turns that observation into an experimental design question: how should mitigation methods be interpreted when class frequency and inherent class difficulty can be varied deliberately?

TCGA-UT is a useful test bed for this question because it provides both slide labels and patch-level evidence derived from diagnostic whole-slide images in The Cancer Genome Atlas (Tomczak et al., 2015; Komura et al., 2022). The native dataset already has a long-tailed slide distribution over 32 cancer types. Native imbalance is realistic, but it entangles prevalence with many other factors: tumor morphology, tissue diversity, slide acquisition, and the representation geometry induced by the frozen encoder. A controlled construction complements the native benchmark by imposing known training-support profiles while holding validation and test data fixed.

The goal is not to introduce a new mitigation method. Instead, the report evaluates representative methods from the companion taxonomy under constructed training distributions. Patch-level methods include cross-entropy, class weighting, balanced sampling, focal loss, and Scholz-style metric losses (Scholz et al., 2024). WSI-bag methods include MIL cross-entropy, class weighting, balanced sampling, focal loss, RankMix, SC-MIL, and MDE-MIL (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025). Patch and WSI-bag methods are reported separately because they consume different prediction units and answer regime-specific questions.

2 Dataset Construction

The construction starts from the same TCGA-UT case-disjoint split protocol as the companion benchmark. For each seed, the validation and test splits are kept fixed, and artificial imbalance is applied only to the training split. The split assignment is patient-disjoint: all slides and feature rows associated with one TCGA participant remain in exactly one of train, validation, or test. This isolates the effect of training support: model selection and held-out evaluation remain comparable across imbalance parameters and mitigation methods without patient leakage.

2.1 Full-Scale Training Pool

All 32 TCGA-UT cancer types are retained. Let $C = 32$ and let $\mathcal{D}_{\text{train}}$ denote the seed-specific training slide pool. For class c , let a_c be the number of available training slides before artificial resampling. The constructed training split must satisfy three constraints:

$$1 \leq n_c \leq a_c, \quad \sum_{c=1}^C n_c = |\mathcal{D}_{\text{train}}|, \quad c \in \{1, \dots, C\}. \quad (1)$$

The first constraint keeps every class represented and prevents oversampling beyond available slides. The second preserves the full training split size. The third makes the benchmark a 32-class TCGA-UT study throughout.

After slide selection, the input evidence is capped in the same way as in the companion benchmark. Patch classifiers use up to 30 deterministic patches per slide. WSI-bag classifiers use up to 30 deterministic frozen feature instances per slide. In both regimes, Virchow2 features remain frozen (Vorontsov et al., 2024; Zimmermann et al., 2024). Features are read from the shared `cls_patchmean` Virchow2 store (`concat(CLS, mean(patch tokens))`), 2,560-dimensional

vectors) used by the companion WSI-bag benchmark. The cap prevents slides with more extracted tumor patches from dominating the training objective while keeping the input budget aligned across regimes.

2.2 Power-Law Training Support

For an ordered class list $(c_1, \dots, c_C)$, the intended share for rank i under imbalance parameter λ is

$$p_i(\lambda) = \frac{i^{-\lambda}}{\sum_{j=1}^C j^{-\lambda}}. \quad (2)$$

When $\lambda = 0$, the intended distribution is uniform over ranks. As λ increases, early-ranked classes receive more training support and late-ranked classes receive less. Raw targets $|\mathcal{D}_{\text{train}}|p_i(\lambda)$ are converted to integer counts by flooring and assigning residual slides by largest fractional remainder.

The target distribution is then projected onto the feasible set in Equation (1). Each class receives one guaranteed slide. Remaining slides are allocated according to Equation (2); classes whose requested support exceeds their available training support are capped, and the unused mass is redistributed over the uncapped classes using the same power-law ranks. This cap-and-redistribute procedure keeps the constructed split full-scale without dropping any cancer type.

2.3 Class Order

The same imbalance severity can have different consequences depending on which classes become frequent. This report therefore uses three class-order regimes. In the *native-prevalence order*, classes are ranked by their TCGA-UT slide support, so the constructed head and tail follow the realistic direction of the native TCGA-UT long tail. In the *easy-to-difficult order*, classes are ranked by decreasing full-support baseline recall from the companion benchmark, averaged across the patch-feature CE and WSI-bag MIL-CE baselines. In the *difficult-to-easy order*, that ranking is reversed. The difficulty orders deliberately couple support with inherent class separability in opposite directions: the first assigns more support to classes that were already easier under balanced/full-support training, whereas the second assigns more support to classes that were difficult despite the frozen Virchow2 representation.

The construction therefore separates support magnitude from the held-out evaluation distribution while also varying the semantic identity of head and tail classes. Frequency is imposed by λ and rank; validation and test cases are unchanged. The companion report’s class-difficulty analysis remains the interpretive background: if a class performs poorly when it is assigned high training support, scarcity is not the only plausible explanation.

3 Methods

The method taxonomy follows the companion benchmark, but this report uses it to study controlled training distributions rather than only the native TCGA-UT distribution. Cross-entropy is the no-mitigation reference. Data-level methods change minibatch exposure through balanced sampling or feature-space bag augmentation. Algorithm-level methods change the per-example loss through class weighting, focal loss, metric-derived objectives, or prototype-affinity objectives. WSI-specific methods change how slide bags are represented or mixed.

Let y be the class label, $\mathbf{z} \in \mathbb{R}^C$ the logits, and p_y the softmax probability of the true class. Cross-entropy optimizes $-\log p_y$. Weighted cross-entropy multiplies this term by a normalized inverse-frequency weight. Focal loss uses

$$\mathcal{L}_{\text{focal}}(\mathbf{z}, y) = -(1 - p_y)^\gamma \log p_y, \quad (3)$$

Regime	Family	Methods	Controlled question
Patch	Baseline and losses	CE, weighted CE, focal loss	Frequency weighting and hard-example emphasis.
Patch	Metric losses	CE + soft F1, CE + soft MCC	Macro-metric losses with balanced minibatch exposure.
Patch	Specialized	CFAL, divide-and-conquer	Prototype-affinity and expert fusion objectives.
WSI bag	Baseline and losses	MIL CE, weighted CE, focal loss	Slide-level analogues of standard losses.
WSI bag	MIL-specific	Balanced sampling, RankMix, SC-MIL, MDE-MIL	Bag-level sampling, augmentation, contrastive structure, and expert training.

Table 1: Method groups evaluated under constructed TCGA-UT training distributions. Patch and WSI-bag methods are compared only within their own input regime.

so confidently classified examples contribute less to the gradient. The soft-F1 and soft-MCC variants add differentiable approximations of macro evaluation metrics to CE, following the imbalance-aware medical-image objectives of Scholz et al. (2024). RankMix, SC-MIL, and MDE-MIL are retained as WSI-bag methods because their mechanisms operate on bags rather than isolated patch embeddings (Chen and Lu, 2023; Juyal et al., 2024; Ling et al., 2025).

4 Evaluation Protocol

Each constructed split is indexed by seed, class-order regime, and imbalance parameter λ . The main reporting points are $\lambda = 0.0$, $\lambda = 1.0$, and $\lambda = 1.3$. For every method and every constructed regime (class order $\times \lambda$), hyperparameters are selected by validation macro F1 across three seeds; ties are broken by validation balanced accuracy. Baseline CE and MIL CE use fixed protocol settings. Selected configurations are evaluated on the fixed test split. Macro F1 is the primary discrimination metric because it weights cancer types equally. Balanced accuracy and accuracy are reported as complementary discrimination metrics. Calibration is summarized with negative log-likelihood, multiclass Brier score, and expected calibration error. For each trained model, a scalar temperature is fitted on the validation probabilities and applied to the corresponding test probabilities, so raw and temperature-scaled calibration can be compared without changing discrimination metrics.

The comparison is intentionally regime-specific. Patch-feature methods are compared only with other patch-feature methods, and WSI-bag methods are compared only with other WSI-bag methods. The two regimes share the case-disjoint splits, frozen Virchow2 representation family, and 30-instance evidence budget, but they differ in prediction unit and aggregation mechanism.

5 Results

The result analysis asks how methods behave across the three class-order regimes and across imbalance severities $\lambda \in \{0.0, 1.0, 1.3\}$. Table 2 and Figure 1 show the imposed support profiles. Tables 3 and 4 compare mitigation methods within the patch-feature and WSI-bag regimes across all nine constructed regimes.

At the patch-feature level, the dominant pattern across all nine constructed regimes is stability: neither changing the class order nor increasing λ produces consistent shifts in balanced accuracy or macro F1 (Table 3). CFAL achieves the highest balanced accuracy of all patch methods (0.763–0.766) and macro F1 (0.766–0.768), with virtually no change across class orders or imbalance severities. The CE-plus-soft-F1 variant follows with balanced accuracy 0.743–

Order	λ	Training slides	Support range
difficult_to_easy	0.0	6110	20.0–567.0
difficult_to_easy	1.0	6110	20.0–567.0
difficult_to_easy	1.3	6110	20.0–567.0
easy_to_difficult	0.0	6110	20.0–567.0
easy_to_difficult	1.0	6110	20.0–567.0
easy_to_difficult	1.3	6110	20.0–567.0
native_prevalence	0.0	6110	20.0–567.0
native_prevalence	1.0	6110	20.0–567.0
native_prevalence	1.3	6110	20.0–567.0

Table 2: Constructed training-support summaries by class-order regime and imbalance severity.

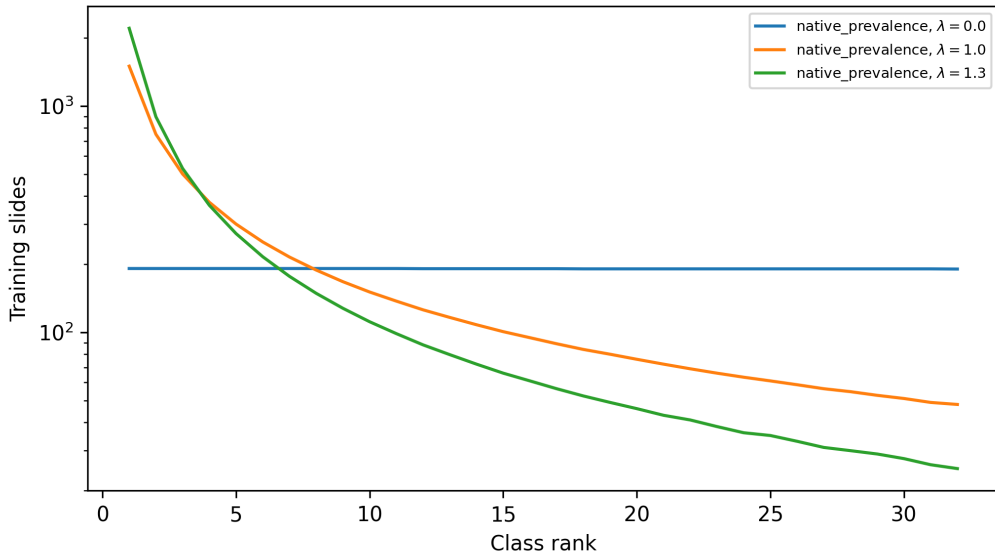


Figure 1: Constructed training support by class rank for the main imbalance settings.

0.759, the widest within-method range of any patch method but still spanning less than two percentage points. Weighted CE, focal loss, and balanced sampling cluster tightly in the 0.740–0.744 balanced-accuracy band. CE-plus-soft-MCC occupies a similar range (0.733–0.751), with somewhat higher variance across seeds at the strongest imbalance. Divide-and-conquer is a consistent outlier at 0.714 balanced accuracy regardless of class order or λ . The absence of systematic rank reversals between methods across the three class-order regimes is the salient patch-level result: altering which classes are placed in the training head, and how aggressively, does not materially change discrimination when frozen Virchow2 patch features are used.

The WSI-bag regime reveals a clearer separation between method families than the patch regime, while also showing stability across class orders (Table 4). MDE-MIL achieves the highest balanced accuracy (0.849–0.866) and macro F1 (0.859–0.870) across all nine constructed regimes. RankMix is the closest competitor, with balanced accuracy 0.850–0.868 and macro F1 0.845–0.852. SC-MIL follows at 0.845–0.854 balanced accuracy, and balanced-sampling MIL reaches 0.840–0.857, with its largest value under native-prevalence ordering at $\lambda = 1.3$. Weighted MIL and focal MIL occupy the lower tier at roughly 0.822–0.852 balanced accuracy, with weighted MIL showing noticeably more variance across seeds at $\lambda = 1.3$. Within each WSI-bag method, performance shifts by at most two percentage points across the three class-order regimes at any fixed λ . The ranking of method families—MDE-MIL and RankMix above SC-MIL above loss-

based variants—is consistent across all nine constructed regimes, confirming that the advantage of WSI-specific bag-composition mechanisms is not contingent on which classes occupy the training head.

6 Discussion

The central finding of the controlled construction is that frozen Virchow2 patch representations absorb most of the variation introduced by different class orderings and imbalance severities. Across all nine constructed regimes, both patch-feature and WSI-bag methods shift by less than two percentage points in balanced accuracy and macro F1. This stability holds even under the difficult-to-easy ordering, which deliberately assigns more training support to the cancer types whose frozen features overlap most with neighbouring classes—exactly the configuration the companion benchmark identified as a risk factor for poor classwise recall. Evidently, the absolute support provided under the full-scale construction is already sufficient for the frozen representation to maintain near-equivalent discrimination regardless of how training mass is distributed across classes.

Patch-feature results cluster tightly across all conditions. CFAL achieves the highest balanced accuracy without dependence on class order or λ ; divide-and-conquer is the weakest patch method throughout. Between these extremes, CE-plus-soft-F1, CE-plus-soft-MCC, weighted CE, focal loss, and balanced sampling occupy a narrow band roughly five percentage points wide. The absence of a strong imbalance effect on patch methods is consistent with the companion report’s observation that a high-quality frozen encoder compresses many patch-level method differences: when features are discriminative, class-frequency adjustments in the loss or sampling schedule have little residual work to do.

WSI-bag methods show more differentiation between families, but their ranking across class orders is stable. MDE-MIL and RankMix maintain their lead over loss-based MIL variants under all three orderings, including the difficult-to-easy regime where one might expect aggregation-based methods to benefit from the larger support given to structurally ambiguous classes. The persistence of this ranking argues that the advantage of WSI-specific bag-composition mechanisms reflects the bag-level aggregation problem in general, not a specific alignment between method design and distribution shape.

Together, these results extend the companion benchmark’s conclusion in a controlled direction. The companion showed that class frequency is not the whole story for classwise performance in the native TCGA-UT distribution. The present experiment shows that when frequency is imposed directly, the constructed support profile has minimal effect on aggregate discrimination for either prediction unit. The prediction-unit dimension—patch versus WSI-bag—continues to dominate method comparisons: the gap between regimes exceeds the within-regime variation due to class order or λ by a large margin.

7 Conclusion

This report defines a full-scale controlled TCGA-UT imbalance benchmark that keeps all 32 cancer types, preserves patient-disjoint validation and test splits, and varies the training support profile under three class-order regimes—native prevalence, easy-to-difficult, and difficult-to-easy—each at imbalance severities $\lambda \in \{0.0, 1.0, 1.3\}$. The primary finding across all nine constructed regimes is robustness: discrimination metrics for both patch-feature and WSI-bag methods shift by less than two percentage points as class order and λ vary, indicating that frozen Virchow2 patch representations contain sufficient class-discriminative structure to maintain near-equivalent performance regardless of the constructed training support profile. At the patch level, CFAL achieves the highest balanced accuracy with no regime dependence;

divide-and-conquer is the weakest patch method throughout. At the WSI-bag level, MDE-MIL and RankMix consistently outperform loss-based MIL variants under all three class-order regimes, an advantage that reflects bag-level aggregation mechanisms rather than alignment with a particular distribution shape. The regime-invariant ranking of WSI-bag method families, combined with the overall patch-level insensitivity to class order, suggests that the prediction unit is a stronger determinant of method utility than the imposed training distribution under this full-scale, frozen-feature setting.

Method	Accuracy	Balanced accuracy	Macro F1
patch_feature_balanced_sampler_ce (difficult_to_easy, $\lambda = 0.0$)	0.801 $\pm$ 0.010	0.742 $\pm$ 0.023	0.738 $\pm$ 0.011
patch_feature_balanced_sampler_ce (difficult_to_easy, $\lambda = 1.0$)	0.798 $\pm$ 0.010	0.740 $\pm$ 0.011	0.739 $\pm$ 0.008
patch_feature_balanced_sampler_ce (difficult_to_easy, $\lambda = 1.3$)	0.801 $\pm$ 0.014	0.744 $\pm$ 0.016	0.741 $\pm$ 0.016
patch_feature_balanced_sampler_ce (easy_to_difficult, $\lambda = 0.0$)	0.795 $\pm$ 0.013	0.739 $\pm$ 0.008	0.729 $\pm$ 0.010
patch_feature_balanced_sampler_ce (easy_to_difficult, $\lambda = 1.0$)	0.802 $\pm$ 0.014	0.742 $\pm$ 0.012	0.739 $\pm$ 0.010
patch_feature_balanced_sampler_ce (easy_to_difficult, $\lambda = 1.3$)	0.799 $\pm$ 0.015	0.744 $\pm$ 0.017	0.740 $\pm$ 0.017
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 0.0$)	0.799 $\pm$ 0.015	0.734 $\pm$ 0.014	0.733 $\pm$ 0.010
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.0$)	0.801 $\pm$ 0.014	0.743 $\pm$ 0.015	0.742 $\pm$ 0.009
patch_feature_balanced_sampler_ce (native_prevalence, $\lambda = 1.3$)	0.799 $\pm$ 0.017	0.739 $\pm$ 0.009	0.738 $\pm$ 0.013
patch_feature_ce_soft_f1_balanced (difficult_to_easy, $\lambda = 0.0$)	0.807 $\pm$ 0.015	0.758 $\pm$ 0.018	0.746 $\pm$ 0.019
patch_feature_ce_soft_f1_balanced (difficult_to_easy, $\lambda = 1.0$)	0.802 $\pm$ 0.013	0.756 $\pm$ 0.006	0.742 $\pm$ 0.011
patch_feature_ce_soft_f1_balanced (difficult_to_easy, $\lambda = 1.3$)	0.797 $\pm$ 0.019	0.751 $\pm$ 0.013	0.738 $\pm$ 0.012
patch_feature_ce_soft_f1_balanced (easy_to_difficult, $\lambda = 0.0$)	0.802 $\pm$ 0.015	0.757 $\pm$ 0.006	0.744 $\pm$ 0.009
patch_feature_ce_soft_f1_balanced (easy_to_difficult, $\lambda = 1.0$)	0.800 $\pm$ 0.010	0.745 $\pm$ 0.003	0.742 $\pm$ 0.006
patch_feature_ce_soft_f1_balanced (easy_to_difficult, $\lambda = 1.3$)	0.801 $\pm$ 0.011	0.743 $\pm$ 0.002	0.740 $\pm$ 0.008
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 0.0$)	0.805 $\pm$ 0.007	0.759 $\pm$ 0.012	0.747 $\pm$ 0.010
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.0$)	0.798 $\pm$ 0.017	0.752 $\pm$ 0.017	0.735 $\pm$ 0.014
patch_feature_ce_soft_f1_balanced (native_prevalence, $\lambda = 1.3$)	0.800 $\pm$ 0.008	0.751 $\pm$ 0.008	0.743 $\pm$ 0.008
patch_feature_ce_soft_mcc_balanced (difficult_to_easy, $\lambda = 0.0$)	0.796 $\pm$ 0.014	0.738 $\pm$ 0.013	0.735 $\pm$ 0.009
patch_feature_ce_soft_mcc_balanced (difficult_to_easy, $\lambda = 1.0$)	0.788 $\pm$ 0.019	0.733 $\pm$ 0.015	0.729 $\pm$ 0.019
patch_feature_ce_soft_mcc_balanced (difficult_to_easy, $\lambda = 1.3$)	0.792 $\pm$ 0.013	0.742 $\pm$ 0.012	0.734 $\pm$ 0.013
patch_feature_ce_soft_mcc_balanced (easy_to_difficult, $\lambda = 0.0$)	0.795 $\pm$ 0.011	0.745 $\pm$ 0.009	0.734 $\pm$ 0.009
patch_feature_ce_soft_mcc_balanced (easy_to_difficult, $\lambda = 1.0$)	0.794 $\pm$ 0.018	0.746 $\pm$ 0.021	0.741 $\pm$ 0.019
patch_feature_ce_soft_mcc_balanced (easy_to_difficult, $\lambda = 1.3$)	0.796 $\pm$ 0.007	0.738 $\pm$ 0.018	0.733 $\pm$ 0.011
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 0.0$)	0.794 $\pm$ 0.018	0.744 $\pm$ 0.015	0.740 $\pm$ 0.019
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.0$)	0.800 $\pm$ 0.012	0.751 $\pm$ 0.014	0.746 $\pm$ 0.017
patch_feature_ce_soft_mcc_balanced (native_prevalence, $\lambda = 1.3$)	0.796 $\pm$ 0.011	0.748 $\pm$ 0.002	0.738 $\pm$ 0.008
patch_feature_cfal (difficult_to_easy, $\lambda = 0.0$)	0.827 $\pm$ 0.010	0.763 $\pm$ 0.006	0.766 $\pm$ 0.005
patch_feature_cfal (difficult_to_easy, $\lambda = 1.0$)	0.827 $\pm$ 0.010	0.763 $\pm$ 0.006	0.766 $\pm$ 0.005
patch_feature_cfal (difficult_to_easy, $\lambda = 1.3$)	0.828 $\pm$ 0.009	0.766 $\pm$ 0.009	0.768 $\pm$ 0.006
patch_feature_cfal (easy_to_difficult, $\lambda = 0.0$)	0.828 $\pm$ 0.009	0.766 $\pm$ 0.009	0.768 $\pm$ 0.006
patch_feature_cfal (easy_to_difficult, $\lambda = 1.0$)	0.827 $\pm$ 0.010	0.763 $\pm$ 0.006	0.766 $\pm$ 0.005
patch_feature_cfal (easy_to_difficult, $\lambda = 1.3$)	0.827 $\pm$ 0.010	0.763 $\pm$ 0.006	0.766 $\pm$ 0.005
patch_feature_cfal (native_prevalence, $\lambda = 0.0$)	0.827 $\pm$ 0.010	0.763 $\pm$ 0.006	0.766 $\pm$ 0.005
patch_feature_cfal (native_prevalence, $\lambda = 1.0$)	0.827 $\pm$ 0.010	0.763 $\pm$ 0.006	0.766 $\pm$ 0.005
patch_feature_cfal (native_prevalence, $\lambda = 1.3$)	0.828 $\pm$ 0.009	0.766 $\pm$ 0.009	0.768 $\pm$ 0.006
patch_feature_divide_conquer (difficult_to_easy, $\lambda = 0.0$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (difficult_to_easy, $\lambda = 1.0$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (difficult_to_easy, $\lambda = 1.3$)	0.792 $\pm$ 0.016	0.714 $\pm$ 0.017	0.725 $\pm$ 0.017
patch_feature_divide_conquer (easy_to_difficult, $\lambda = 0.0$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (easy_to_difficult, $\lambda = 1.0$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (easy_to_difficult, $\lambda = 1.3$)	0.792 $\pm$ 0.014	0.713 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (native_prevalence, $\lambda = 0.0$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (native_prevalence, $\lambda = 1.0$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_divide_conquer (native_prevalence, $\lambda = 1.3$)	0.792 $\pm$ 0.015	0.714 $\pm$ 0.011	0.725 $\pm$ 0.012
patch_feature_focal (difficult_to_easy, $\lambda = 0.0$)	0.804 $\pm$ 0.015	0.741 $\pm$ 0.010	0.738 $\pm$ 0.013
patch_feature_focal (difficult_to_easy, $\lambda = 1.0$)	0.807 $\pm$ 0.013	0.744 $\pm$ 0.014	0.742 $\pm$ 0.015
patch_feature_focal (difficult_to_easy, $\lambda = 1.3$)	0.807 $\pm$ 0.013	0.744 $\pm$ 0.014	0.742 $\pm$ 0.015
patch_feature_focal (easy_to_difficult, $\lambda = 0.0$)	0.804 $\pm$ 0.015	0.741 $\pm$ 0.010	0.738 $\pm$ 0.013
patch_feature_focal (easy_to_difficult, $\lambda = 1.0$)	0.807 $\pm$ 0.013	0.744 $\pm$ 0.014	0.742 $\pm$ 0.015
patch_feature_focal (easy_to_difficult, $\lambda = 1.3$)	0.804 $\pm$ 0.015	0.741 $\pm$ 0.010	0.738 $\pm$ 0.013
patch_feature_focal (native_prevalence, $\lambda = 0.0$)	0.807 $\pm$ 0.013	0.744 $\pm$ 0.014	0.742 $\pm$ 0.015
patch_feature_focal (native_prevalence, $\lambda = 1.0$)	0.804 $\pm$ 0.015	0.741 $\pm$ 0.010	0.738 $\pm$ 0.013
patch_feature_focal (native_prevalence, $\lambda = 1.3$)	0.807 $\pm$ 0.013	0.744 $\pm$ 0.014	0.742 $\pm$ 0.015
patch_feature_weighted_ce (difficult_to_easy, $\lambda = 0.0$)	0.808 $\pm$ 0.015	0.743 $\pm$ 0.011	0.744 $\pm$ 0.011
patch_feature_weighted_ce (difficult_to_easy, $\lambda = 1.0$)	0.808 $\pm$ 0.015	0.743 $\pm$ 0.011	0.744 $\pm$ 0.011
patch_feature_weighted_ce (difficult_to_easy, $\lambda = 1.3$)	0.805 $\pm$ 0.017	0.740 $\pm$ 0.008	0.742 $\pm$ 0.009
patch_feature_weighted_ce (easy_to_difficult, $\lambda = 0.0$)	0.804 $\pm$ 0.010	0.741 $\pm$ 0.014	0.742 $\pm$ 0.008
patch_feature_weighted_ce (easy_to_difficult, $\lambda = 1.0$)	0.804 $\pm$ 0.010	0.741 $\pm$ 0.014	0.742 $\pm$ 0.008
patch_feature_weighted_ce (easy_to_difficult, $\lambda = 1.3$)	0.804 $\pm$ 0.010	0.741 $\pm$ 0.014	0.742 $\pm$ 0.008
patch_feature_weighted_ce (native_prevalence, $\lambda = 0.0$)	0.805 $\pm$ 0.017	0.740 $\pm$ 0.008	0.742 $\pm$ 0.009
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.0$)	0.805 $\pm$ 0.017	0.740 $\pm$ 0.008	0.742 $\pm$ 0.009
patch_feature_weighted_ce (native_prevalence, $\lambda = 1.3$)	0.804 $\pm$ 0.010	0.741 $\pm$ 0.014	0.742 $\pm$ 0.008

Table 3: Patch-feature test results for validation-selected configurations under constructed training distributions.

Method	Accuracy	Balanced accuracy	Macro F1
mde_mil (difficult_to_easy, $\lambda = 0.0$)	0.894 ± 0.007	0.858 ± 0.003	0.862 ± 0.003
mde_mil (difficult_to_easy, $\lambda = 1.0$)	0.894 ± 0.007	0.849 ± 0.003	0.859 ± 0.009
mde_mil (difficult_to_easy, $\lambda = 1.3$)	0.898 ± 0.007	0.858 ± 0.008	0.864 ± 0.007
mde_mil (easy_to_difficult, $\lambda = 0.0$)	0.900 ± 0.009	0.866 ± 0.010	0.870 ± 0.012
mde_mil (easy_to_difficult, $\lambda = 1.0$)	0.900 ± 0.008	0.860 ± 0.006	0.866 ± 0.006
mde_mil (easy_to_difficult, $\lambda = 1.3$)	0.894 ± 0.007	0.858 ± 0.003	0.862 ± 0.003
mde_mil (native_prevalence, $\lambda = 0.0$)	0.896 ± 0.011	0.862 ± 0.006	0.866 ± 0.010
mde_mil (native_prevalence, $\lambda = 1.0$)	0.894 ± 0.007	0.855 ± 0.007	0.859 ± 0.003
mde_mil (native_prevalence, $\lambda = 1.3$)	0.900 ± 0.009	0.866 ± 0.010	0.870 ± 0.012
mil_balanced_sampler.ce (difficult_to_easy, $\lambda = 0.0$)	0.878 ± 0.006	0.849 ± 0.010	0.851 ± 0.011
mil_balanced_sampler.ce (difficult_to_easy, $\lambda = 1.0$)	0.871 ± 0.009	0.846 ± 0.013	0.844 ± 0.013
mil_balanced_sampler.ce (difficult_to_easy, $\lambda = 1.3$)	0.876 ± 0.010	0.840 ± 0.004	0.836 ± 0.010
mil_balanced_sampler.ce (easy_to_difficult, $\lambda = 0.0$)	0.878 ± 0.006	0.849 ± 0.010	0.851 ± 0.011
mil_balanced_sampler.ce (easy_to_difficult, $\lambda = 1.0$)	0.878 ± 0.006	0.849 ± 0.010	0.851 ± 0.011
mil_balanced_sampler.ce (easy_to_difficult, $\lambda = 1.3$)	0.877 ± 0.005	0.846 ± 0.006	0.847 ± 0.004
mil_balanced_sampler.ce (native_prevalence, $\lambda = 0.0$)	0.879 ± 0.001	0.847 ± 0.004	0.843 ± 0.009
mil_balanced_sampler.ce (native_prevalence, $\lambda = 1.0$)	0.876 ± 0.010	0.840 ± 0.004	0.836 ± 0.010
mil_balanced_sampler.ce (native_prevalence, $\lambda = 1.3$)	0.880 ± 0.010	0.857 ± 0.023	0.855 ± 0.018
mil_focal (difficult_to_easy, $\lambda = 0.0$)	0.879 ± 0.006	0.842 ± 0.008	0.839 ± 0.009
mil_focal (difficult_to_easy, $\lambda = 1.0$)	0.881 ± 0.011	0.841 ± 0.008	0.842 ± 0.007
mil_focal (difficult_to_easy, $\lambda = 1.3$)	0.883 ± 0.008	0.846 ± 0.008	0.848 ± 0.010
mil_focal (easy_to_difficult, $\lambda = 0.0$)	0.878 ± 0.003	0.826 ± 0.009	0.831 ± 0.004
mil_focal (easy_to_difficult, $\lambda = 1.0$)	0.881 ± 0.006	0.838 ± 0.009	0.842 ± 0.004
mil_focal (easy_to_difficult, $\lambda = 1.3$)	0.877 ± 0.003	0.830 ± 0.009	0.835 ± 0.007
mil_focal (native_prevalence, $\lambda = 0.0$)	0.878 ± 0.003	0.826 ± 0.009	0.831 ± 0.004
mil_focal (native_prevalence, $\lambda = 1.0$)	0.872 ± 0.008	0.822 ± 0.004	0.824 ± 0.011
mil_focal (native_prevalence, $\lambda = 1.3$)	0.875 ± 0.004	0.828 ± 0.009	0.828 ± 0.011
mil_weighted.ce (difficult_to_easy, $\lambda = 0.0$)	0.876 ± 0.006	0.850 ± 0.007	0.836 ± 0.017
mil_weighted.ce (difficult_to_easy, $\lambda = 1.0$)	0.880 ± 0.005	0.852 ± 0.013	0.847 ± 0.010
mil_weighted.ce (difficult_to_easy, $\lambda = 1.3$)	0.872 ± 0.013	0.825 ± 0.014	0.826 ± 0.013
mil_weighted.ce (easy_to_difficult, $\lambda = 0.0$)	0.881 ± 0.007	0.845 ± 0.020	0.846 ± 0.017
mil_weighted.ce (easy_to_difficult, $\lambda = 1.0$)	0.883 ± 0.009	0.843 ± 0.017	0.843 ± 0.012
mil_weighted.ce (easy_to_difficult, $\lambda = 1.3$)	0.874 ± 0.013	0.832 ± 0.022	0.830 ± 0.019
mil_weighted.ce (native_prevalence, $\lambda = 0.0$)	0.878 ± 0.009	0.842 ± 0.016	0.839 ± 0.010
mil_weighted.ce (native_prevalence, $\lambda = 1.0$)	0.882 ± 0.019	0.847 ± 0.017	0.847 ± 0.021
mil_weighted.ce (native_prevalence, $\lambda = 1.3$)	0.874 ± 0.013	0.832 ± 0.022	0.830 ± 0.019
rankmix_mil (difficult_to_easy, $\lambda = 0.0$)	0.890 ± 0.006	0.859 ± 0.010	0.846 ± 0.011
rankmix_mil (difficult_to_easy, $\lambda = 1.0$)	0.894 ± 0.003	0.862 ± 0.008	0.852 ± 0.007
rankmix_mil (difficult_to_easy, $\lambda = 1.3$)	0.889 ± 0.010	0.850 ± 0.010	0.845 ± 0.014
rankmix_mil (easy_to_difficult, $\lambda = 0.0$)	0.892 ± 0.005	0.855 ± 0.012	0.851 ± 0.013
rankmix_mil (easy_to_difficult, $\lambda = 1.0$)	0.892 ± 0.004	0.852 ± 0.010	0.848 ± 0.013
rankmix_mil (easy_to_difficult, $\lambda = 1.3$)	0.892 ± 0.004	0.852 ± 0.010	0.848 ± 0.013
rankmix_mil (native_prevalence, $\lambda = 0.0$)	0.892 ± 0.004	0.852 ± 0.010	0.848 ± 0.013
rankmix_mil (native_prevalence, $\lambda = 1.0$)	0.887 ± 0.004	0.868 ± 0.009	0.852 ± 0.011
rankmix_mil (native_prevalence, $\lambda = 1.3$)	0.892 ± 0.004	0.852 ± 0.010	0.848 ± 0.013
sc_mil (difficult_to_easy, $\lambda = 0.0$)	0.881 ± 0.009	0.852 ± 0.020	0.843 ± 0.016
sc_mil (difficult_to_easy, $\lambda = 1.0$)	0.883 ± 0.011	0.853 ± 0.017	0.850 ± 0.021
sc_mil (difficult_to_easy, $\lambda = 1.3$)	0.877 ± 0.011	0.850 ± 0.016	0.848 ± 0.023
sc_mil (easy_to_difficult, $\lambda = 0.0$)	0.883 ± 0.002	0.854 ± 0.008	0.849 ± 0.008
sc_mil (easy_to_difficult, $\lambda = 1.0$)	0.879 ± 0.007	0.845 ± 0.008	0.841 ± 0.014
sc_mil (easy_to_difficult, $\lambda = 1.3$)	0.883 ± 0.002	0.854 ± 0.008	0.849 ± 0.008
sc_mil (native_prevalence, $\lambda = 0.0$)	0.885 ± 0.003	0.847 ± 0.021	0.843 ± 0.017
sc_mil (native_prevalence, $\lambda = 1.0$)	0.882 ± 0.007	0.854 ± 0.018	0.844 ± 0.015
sc_mil (native_prevalence, $\lambda = 1.3$)	0.883 ± 0.002	0.854 ± 0.008	0.849 ± 0.008

Table 4: WSI-bag test results for validation-selected configurations under constructed training distributions.

Method	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
Patch calibration table unavailable.							

Table 5: Patch-feature calibration before and after validation-fitted temperature scaling.

Method	NLL	NLL+TS	Brier	Brier+TS	ECE	ECE+TS	T
WSI-bag calibration table unavailable.							

Table 6: WSI-bag calibration before and after validation-fitted temperature scaling.

References

- Yuan-Chih Chen and Chun-Shien Lu. RankMix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23936–23945, 2023. doi: 10.1109/CVPR52729.2023.02397.
- Dinkar Juyal, Siddhant Shingi, Syed Ashar Javed, Harshith Padigela, Chintan Shah, Anand Sampat, Archit Khosla, John Abel, and Amaro Taylor-Weiner. SC-MIL: Supervised contrastive multiple instance learning for imbalanced classification in pathology. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7016–7025, 2024. doi: 10.1109/WACV57761.2024.00702.
- Daisuke Komura, Akihiro Kawabe, Kenta Fukuta, Kazuya Sano, Tsukasa Umezaki, Haruka Koda, Riku Suzuki, Keisuke Tominaga, Michiaki Ochi, Hirofumi Konishi, et al. Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424, 2022. doi: 10.1016/j.celrep.2022.110424.
- Xitong Ling, Yifeng Ping, Jiawen Li, Jing Peng, and Yuxuan Chen. MDE-MIL. *Dos mil tres mil*, 27, 2025. doi: 10.35707/dostresmil/27.
- Daniel Scholz, Ayhan Can Erdur, Josef Buchner, Jan C. Peeken, Daniel Rueckert, and Benedikt Wiestler. Imbalance-aware loss functions improve medical image classification. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 1341–1356. PMLR, 2024.
- Katarzyna Tomczak, Patrycja Czerwinska, and Maciej Wiznerowicz. The cancer genome atlas (TCGA): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77, 2015. doi: 10.5114/wo.2014.47136.
- Eugene Vorontsov, Alican Bozkurt, Adam Casson, George Shaikovski, Michal Zelechowski, Kristen Severson, Eric Zimmermann, et al. A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935, 2024. doi: 10.1038/s41591-024-03141-0.
- Eric Zimmermann, Eugene Vorontsov, Julian Viret, Adam Casson, Michal Zelechowski, and George Shaikovski. Virchow2. *2024 Shell TechXplorer Digest*, 2024(1), 2024. doi: 10.52196/208316.24.1.